

The Role of AI

Algorithm to bedside symposia





DATA STORIES

From algorithm to bedside

Nermin Ghith, dual B.Sc. in bio/chem.; MPH, PhD

Senior Forsker

Gentofte or Herlev Hospital

03-14 Feb. 2025

Octopus- Senior Epidemiologist

Projects

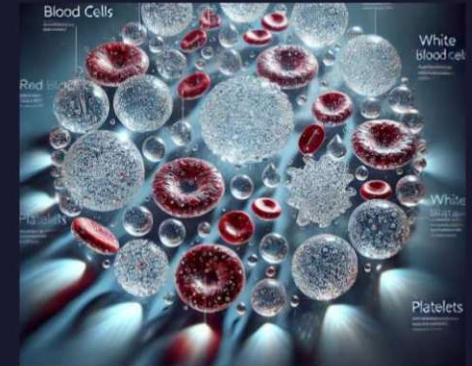
Combine and integrate large-scale data:

- biological samples and data, contextual information (family, institutional, environmental and community-specific effects), risk factors and lifestyle, clinical interventions, treatments, and health outcomes

Model and understand

- The determinants of health and causes of illness in the human population

Nesting a coherent scientific insight to advance clinical care, health policies, and population health!



Data

- Danish and Nordic registries
- WHO national, regional, and global databases, surveys, and questionnaire data
- Industry datasets (e.g, GIDEON, IQVIA)
- Own projects' data (e.g., interviews with patients and healthcare professionals).

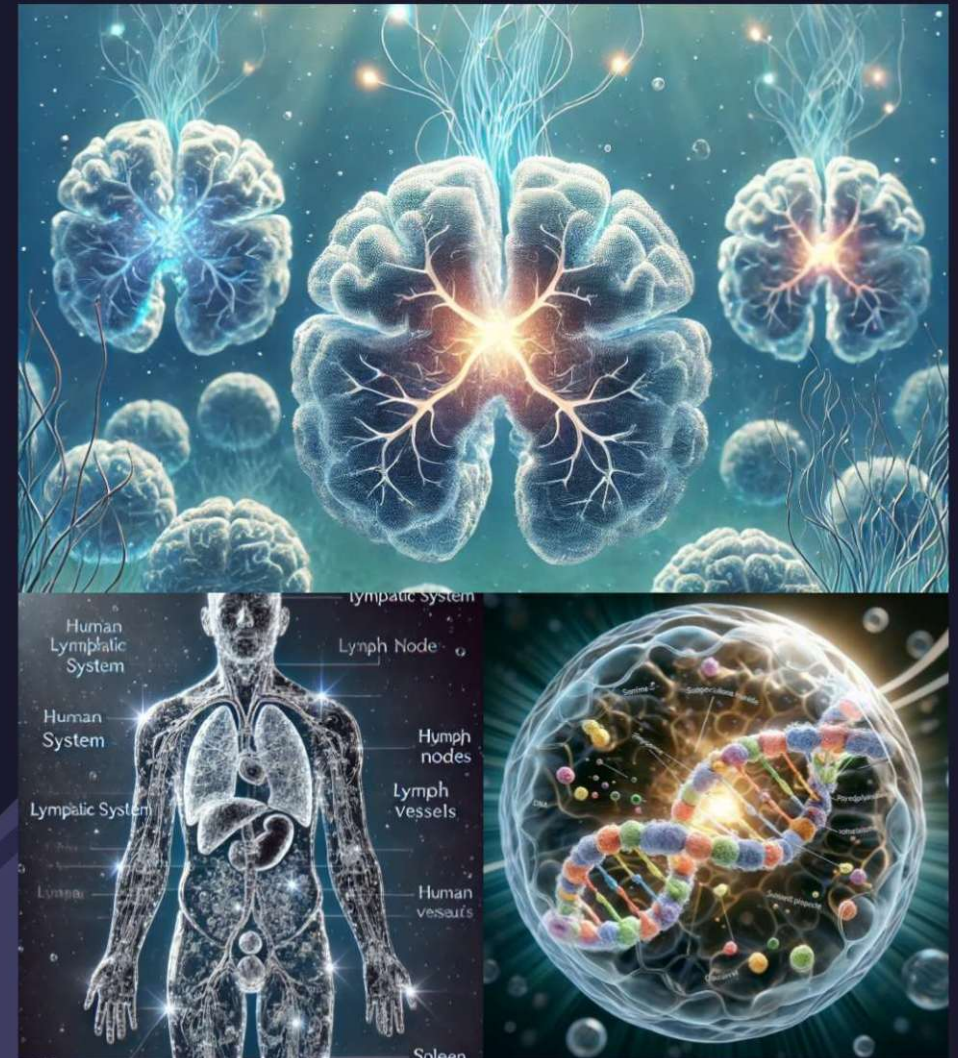


My research and technical areas

Register-based Epidemiology / Health Data Science

(Convergence of Epidemiology, Biostatistics, and Data Science)

- [Late effects, drug use and QoL](#) (SeniorForsker/PostDoc,WHO)
- [\(MetaGenomics/Microbiome\)](#): Global Burden of Antimicrobial resistance and infectious diseases (DTU,WHO)
- [Multimorbidity and polypharmacy](#) (seniorForsker/PostDoc,WHO)
- [Global and Regional Burden of Disease](#) (DTU,WHO)
- [Contextual epidemiology](#), Clinical care, and health outcomes (Ph.D, WHO)
- [Behavioral Risk Factors](#) (e.g., smoking, substance use, diet) – WHO, master thesis



Lunch symposia

Omics-Based Clinical and Population Studies (**session 1: Mon. 3 Feb. 2025** - Specific Case Studies)

Contextual Epidemiology of Cardiometabolic Conditions and Multi-morbidity (**session 2: Thurs. 6 Feb. 2025**)

Scientific Methods in Biomedical Research (**sessions 3 and 4: Mon and Thurs. 10 and 13 Feb. 2025**)

- Omics in biomedical research
- Global Burden of Disease, Health Analytics, and Access to Therapeutics
- Evidence and analytics





Algorithm to bedside

Omics-Based Clinical and Population Studies

(session I-I)

Antimicrobial Resistance : Global burden of AMR – Microbiome and Metagenomics

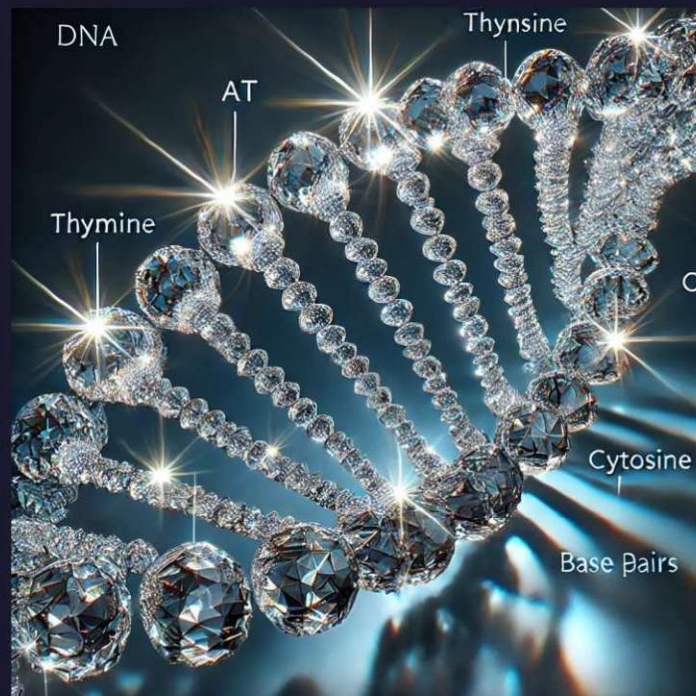


28-01-2025

@NerminGhith



Background



AMR

Inequality in global health outcomes such as antimicrobial resistance (AMR) is shaped by **macro** determinants of health within and between countries.

Aim

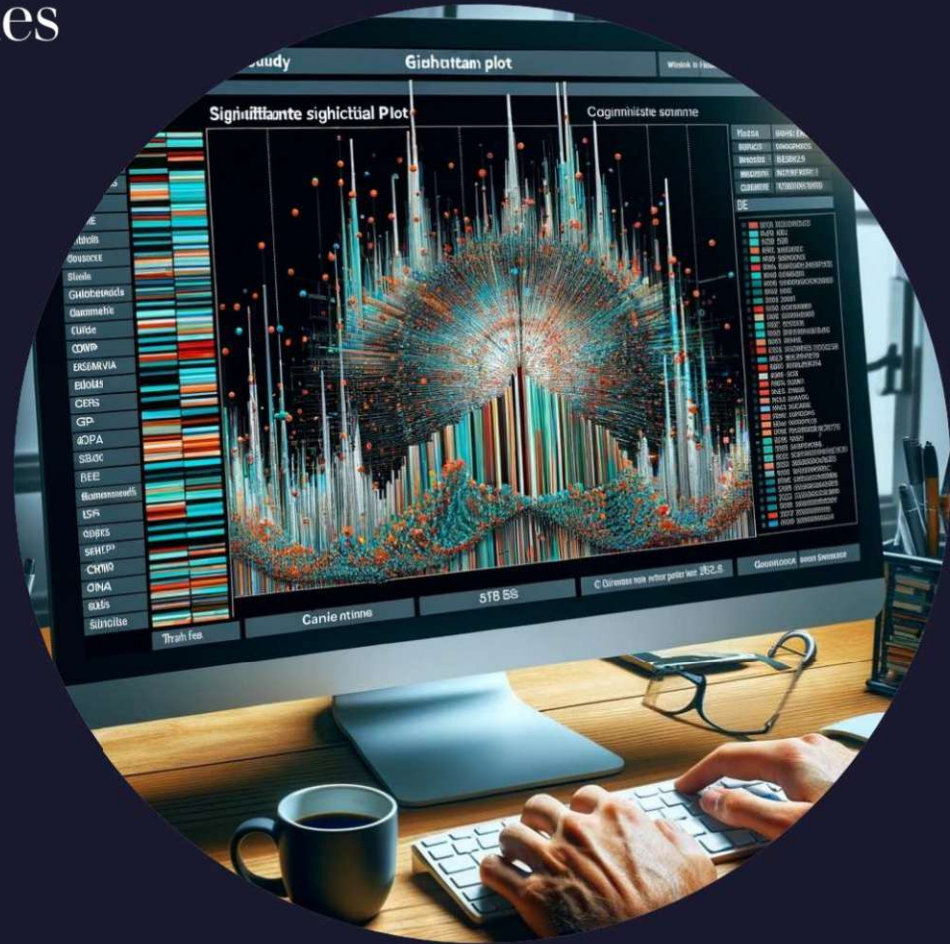
This project aims to assess the potential relationship between the community (at the **city level**) progress towards achieving the sustainability development goals (**SDG**) and the burden of antimicrobial resistance (**AMR**).

Framework

The study identifies **classes** of antimicrobial resistance that varies between the sampling **sites** within **cities**, and compliance with **SDGs** goals that might explain **variation** in the global burden of AMR between countries.

DTU research role (using samples from 104 countries and more than 230 cities)

- Identify **macro** determinants of AMR
- Assess the dual burden of AMR and **infectious** diseases
- Assess the relationship between **morbidity** and **mortality** profiles (i.e., childhood and adult cancer; and cardiometabolic conditions) and the global burden of AMR
- Assess the relationship between **SDGs** and AMR

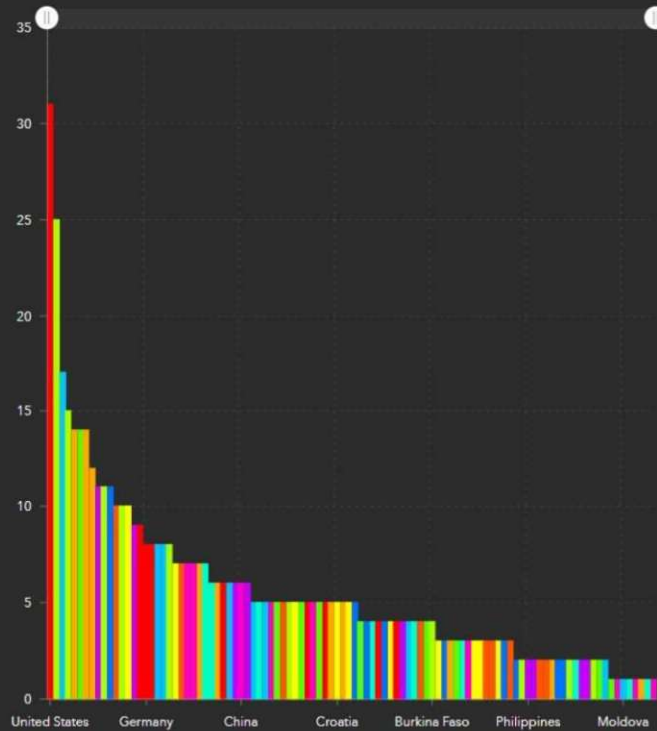


Own data

Global Monitoring of Antimicrobial Resistance using Urban Sewage Samples (Phase II): Global Sewage Surveillance Project (...)

Data measures: CLR values (Resfinder and Bacterial Load) and metadata

Sample per country



Source: Esri, DigitalGlobe, GeoEye, Earthstar Geographics, CNES/Airbus DS, USDA, USGS, AeroGRID, IGN... Powered by Esri

About:

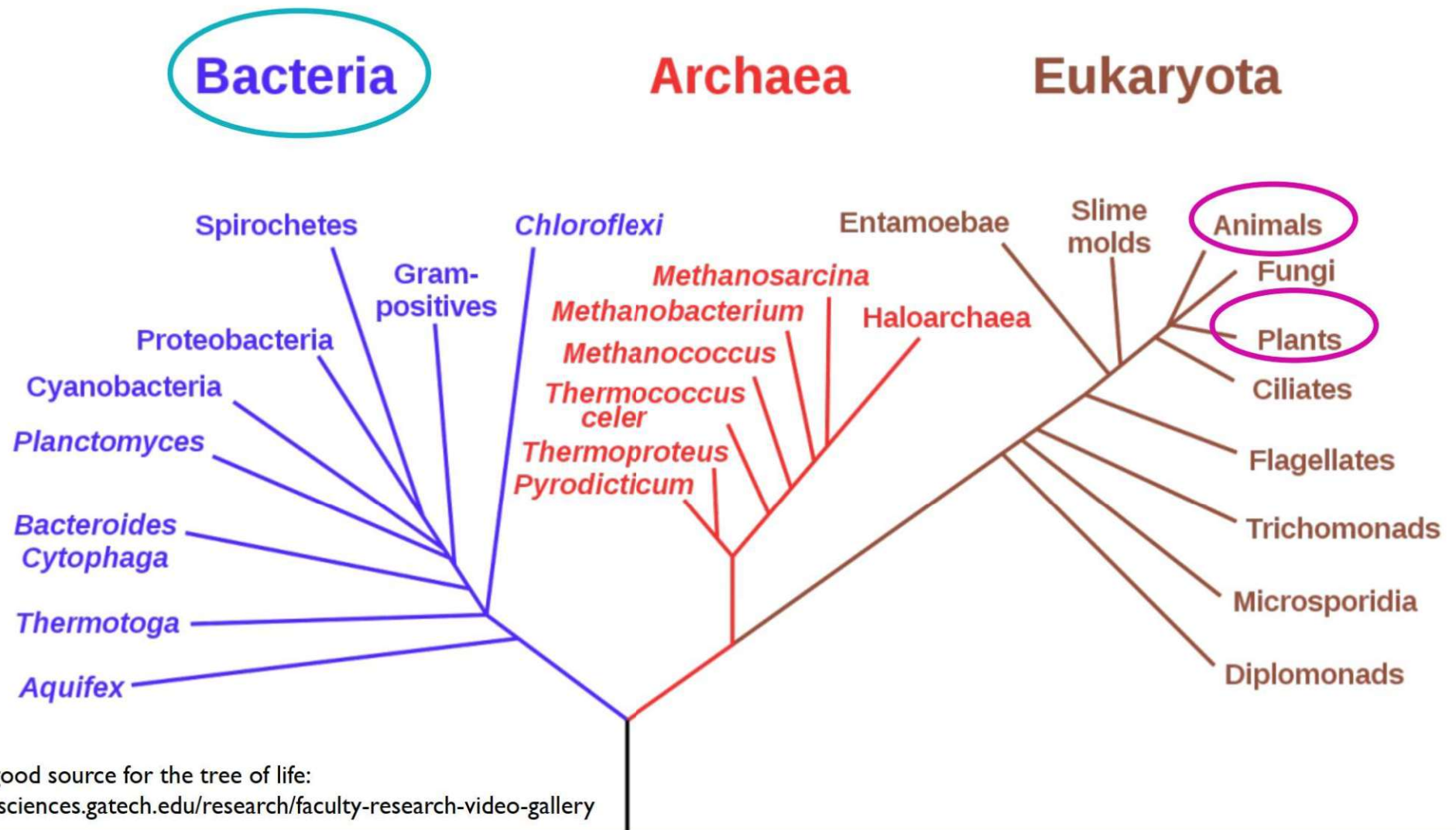
This dashboard was developed by Nermin Ghith, Postdoc Researcher at the Research Group for Genomic Epidemiology, Division for Global Surveillance, National Food Institute, Technical University of Denmark (DTU Food).

Go to project StoryMaps:

Country/City/Municipality/Sample ID

- Albania
Tirana
2016_pil_alb_01
- Albania
Tirana
2017_nov_alb_01
- Argentina
Buenos Aires
2018_nov_arg_01
- Argentina
Buenos Aires
2018_jun_arg_01
- Argentina
Buenos Aires
2017_nov_arg_01
- Australia
Maningrida
2018_nov_aus_02
- Australia
D...

What are microorganisms?



The Oral and Gut Bacterial Microbiomes: Similarities, Differences, and Connections

• July 2020

• Biological Research for Nursing 23(1):109980042094160

• DOI:

• [10.1177/1099800420941606](https://doi.org/10.1177/1099800420941606)

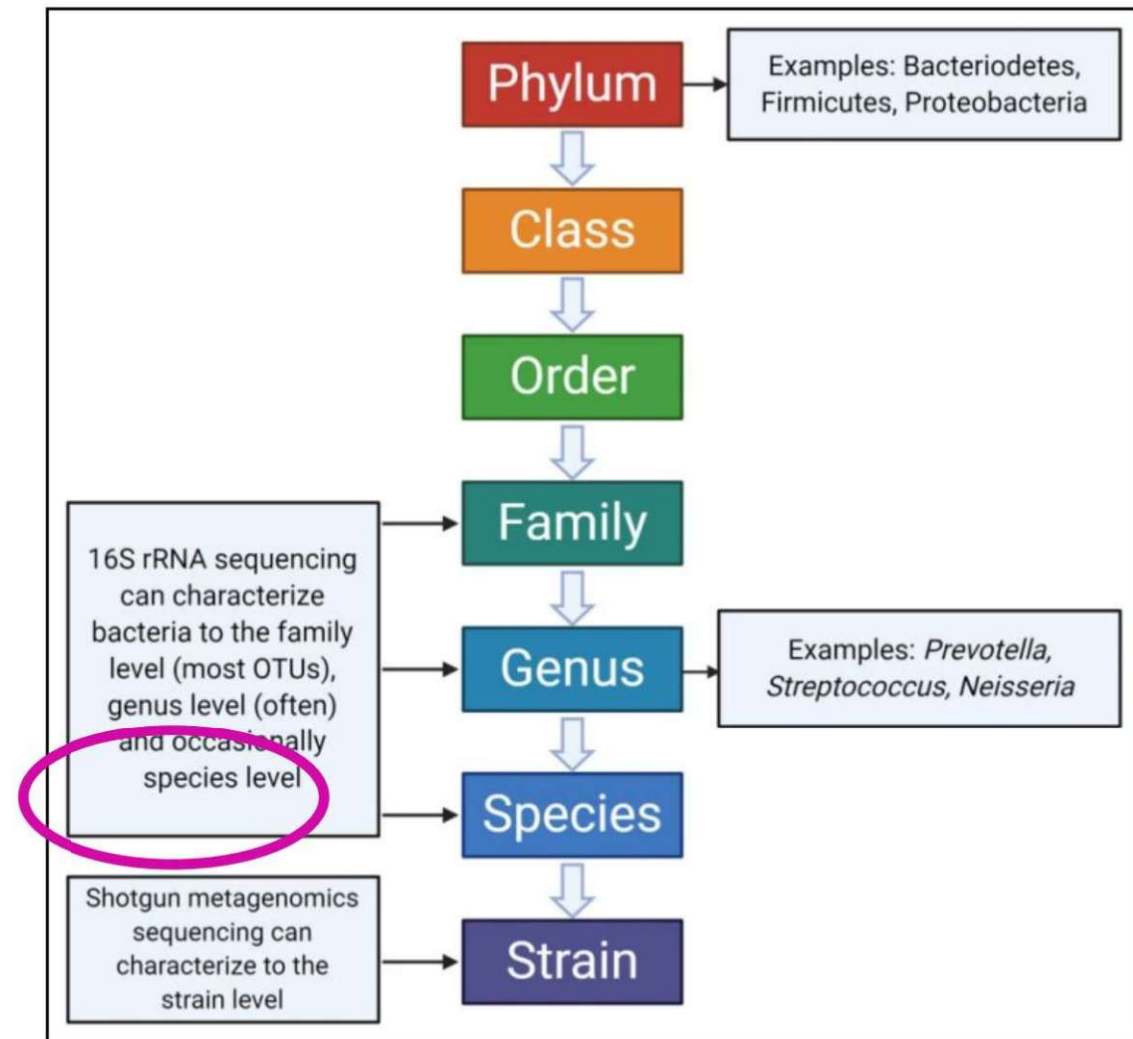
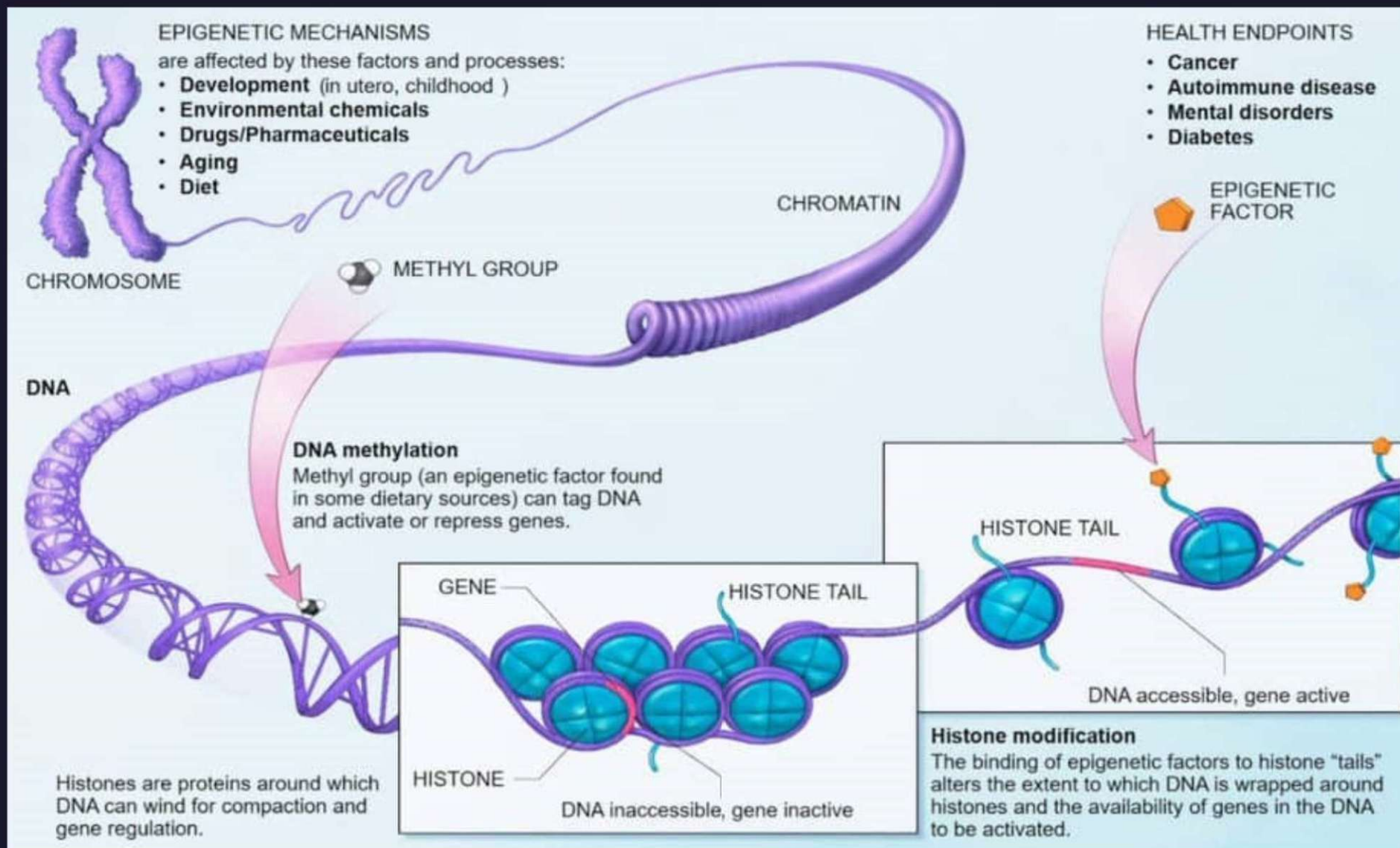
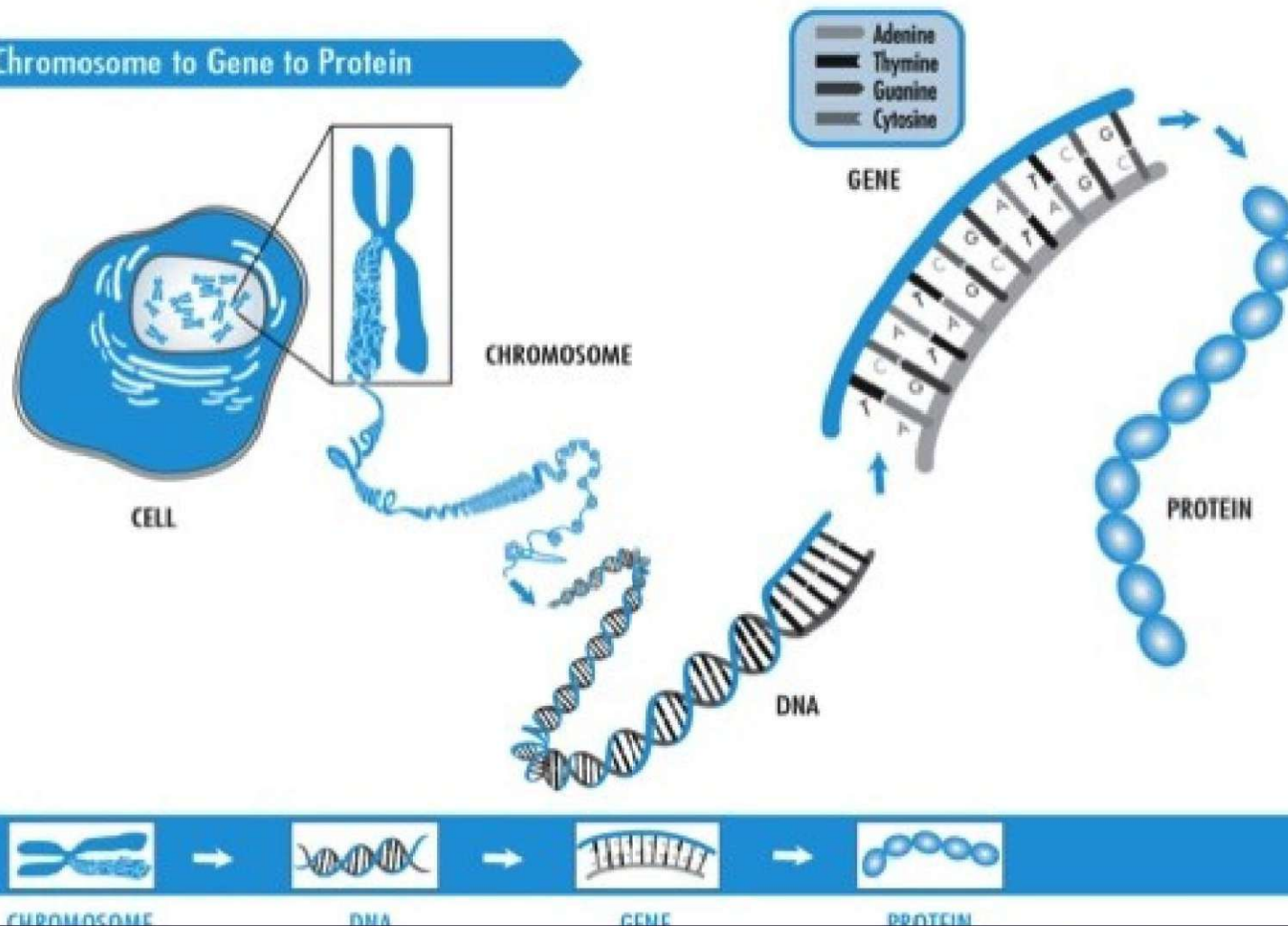


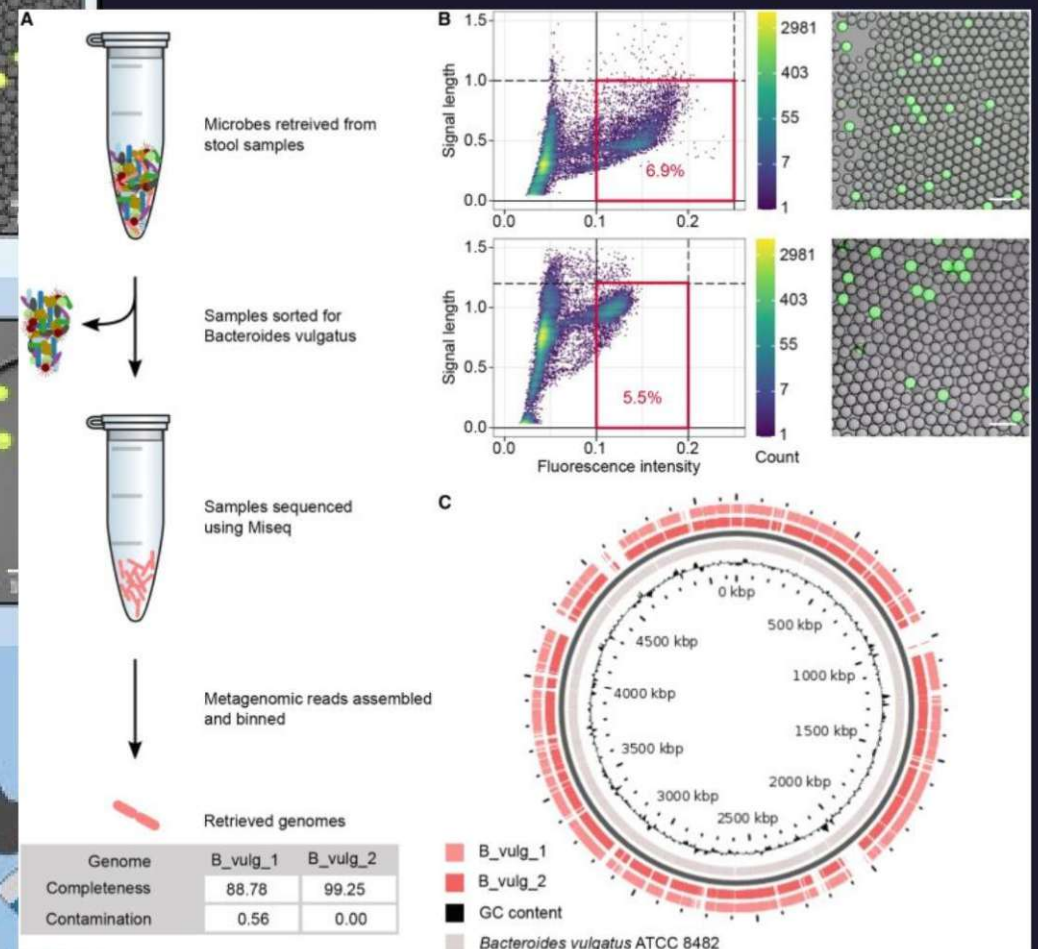
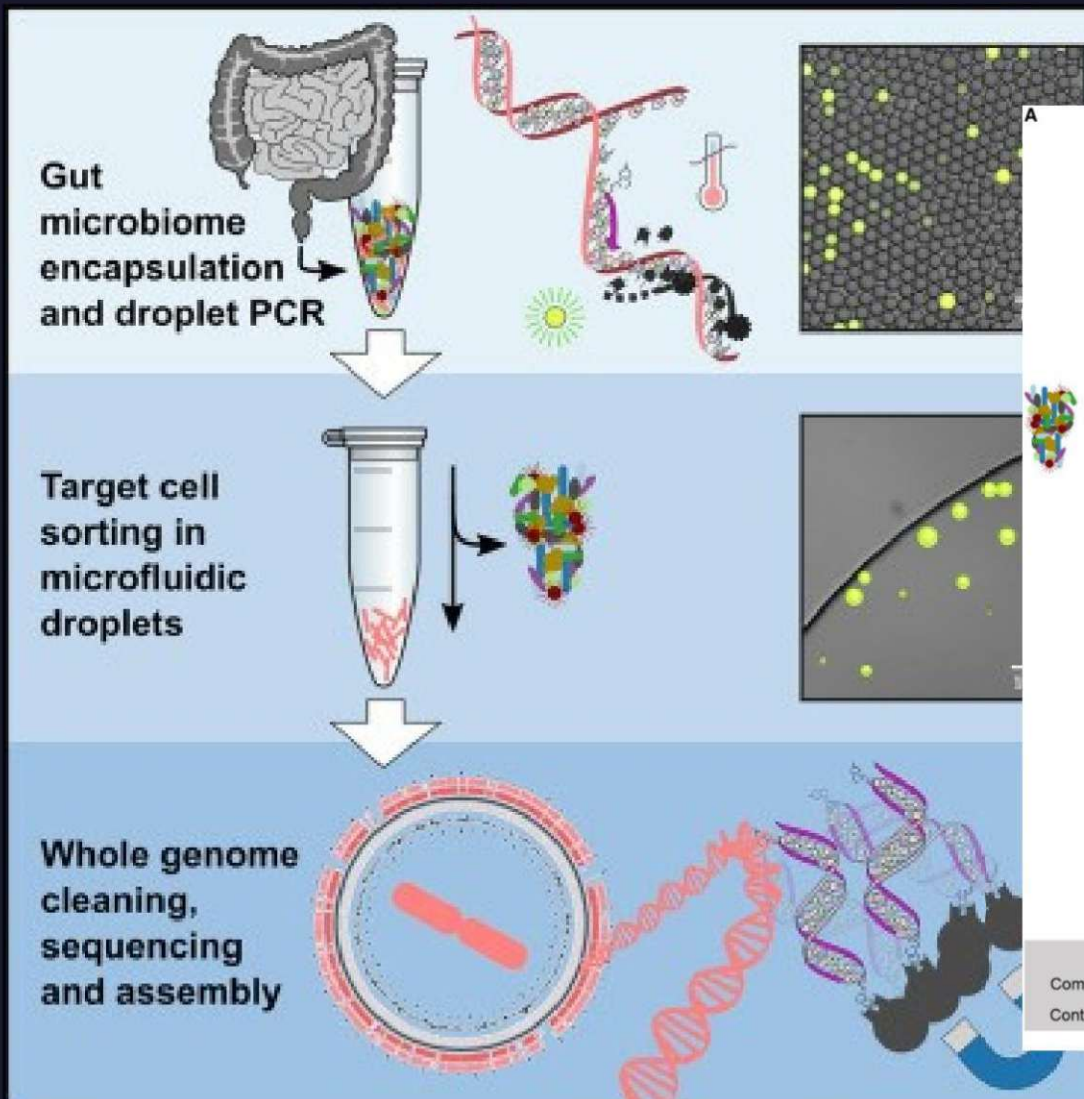
Figure 5. Taxonomic classification of bacteria from phylum to strain level.

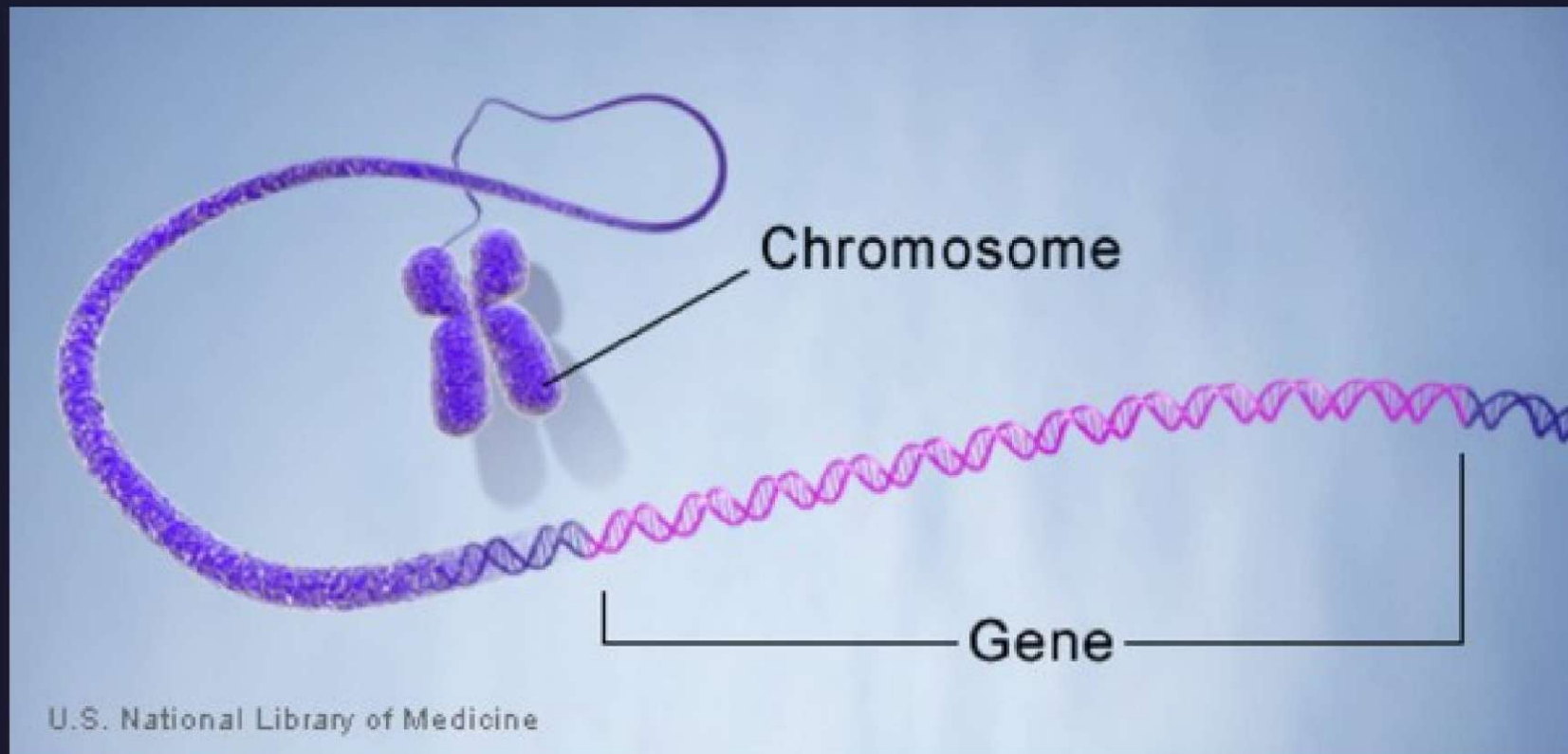


Chromosome to Gene to Protein



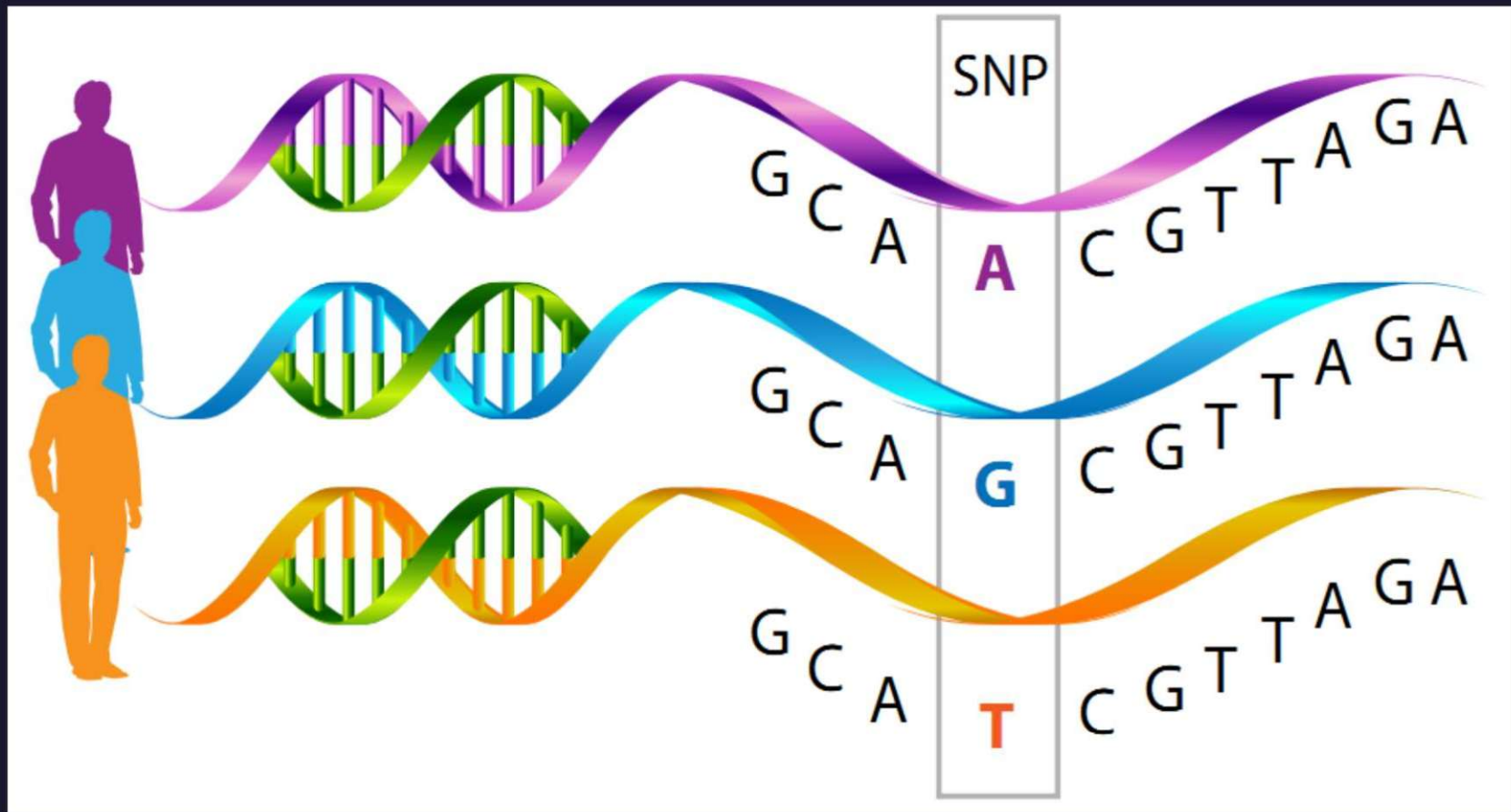
Source: <http://epilepsygenetics.net/2016/05/25/explaining-variants-of-uncertain-significance-a-guide-for-clinicians/>





A gene is the basic physical and functional unit of heredity. Genes are made up of DNA. Some genes act as instructions to make molecules called **proteins**, which are needed for the body to function. However, many genes do not code for proteins, instead they help control other genes.

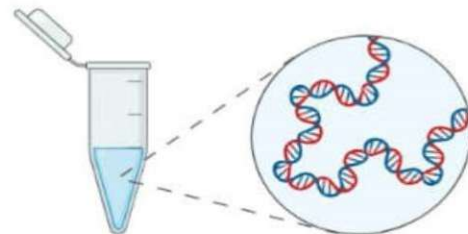
Source: <https://medlineplus.gov/genetics/understanding/basics/gene/>



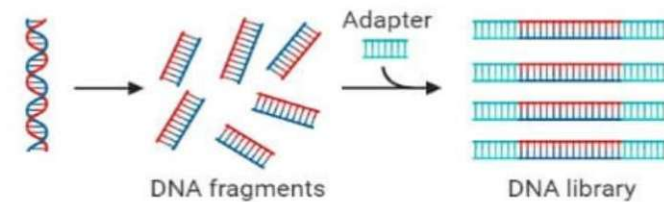
SNPs, or Single Nucleotide Polymorphisms (pronounced snips), are single base changes within genes, akin to single letter spelling variations. (Think analyse vs analyze.) It is important to emphasize that these are variations, NOT mutations. SNPs represent 90+% of the total variation amongst individuals, meaning that nearly every genetic difference you encounter on a daily basis is caused by single nucleotide changes in various genes. These tiny variations (SNPs) may slightly change the function of the protein encoded by the corresponding gene.

Source: <https://www.nutrigeneticsspecialists.com/single-post/2017/03/27/what-is-a-snp>
 28-01-2025 @NerminGhiith

Step 1:
DNA extraction

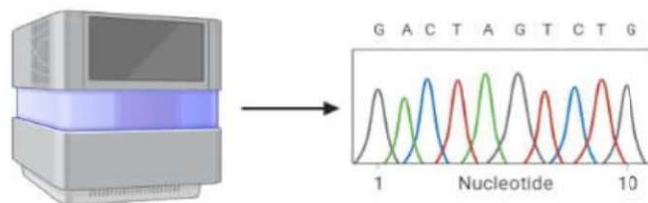


Step 2:
Library preparation

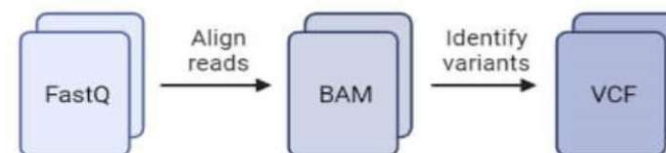


**Next Generation
Sequencing Workflow**

Step 3:
Sequencing



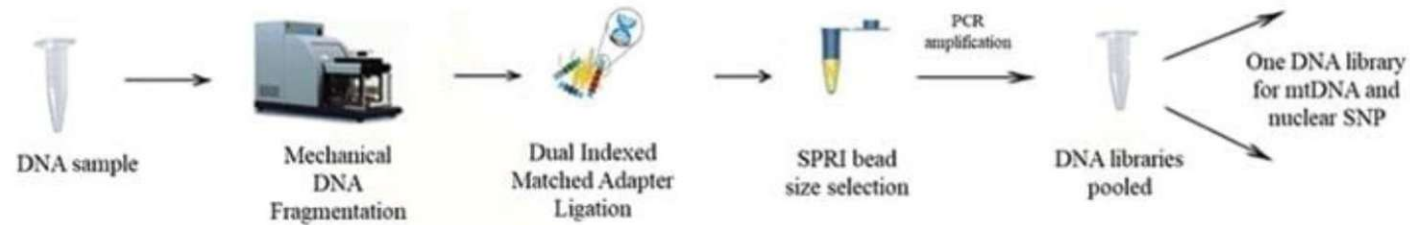
Step 4:
Analysis



NGS is a powerful technology that has transformed the health data landscape. Among many other benefits, it has drastically improved our ability to determine the presence of AMR genes (bacterial genes known to confer resistance to an antimicrobial drug)

Source: <https://microbenotes.com/next-generation-sequencing-ngs/>

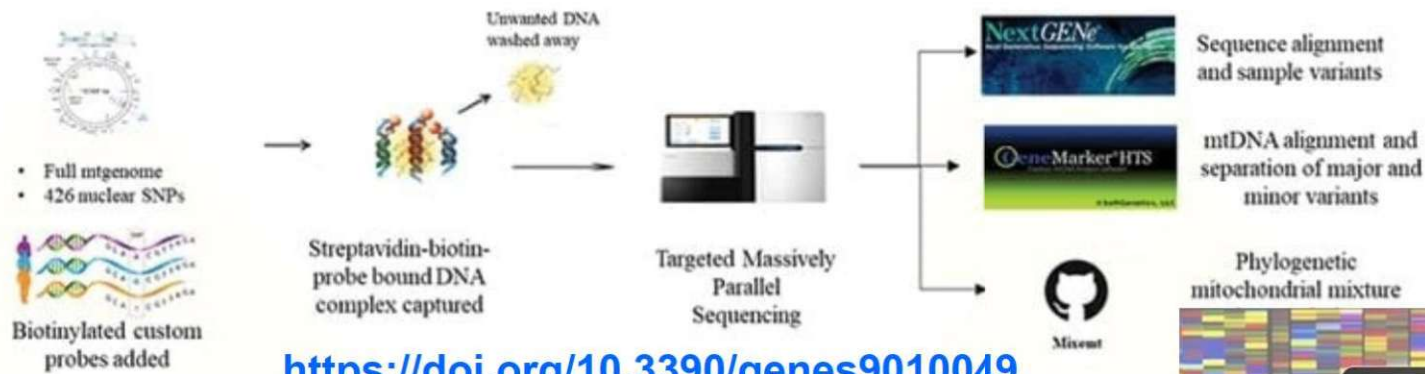
1) DNA Fragmentation and Library Preparation



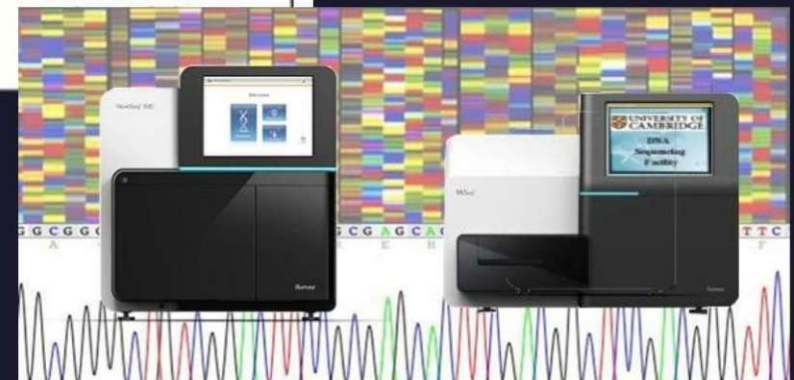
2) Probe Capture and Enrichment

3) Next Gen Sequencing

4) Data Analysis



<https://doi.org/10.3390/genes9010049>



Source: University of Cambridge, Research Facilities at the Department of Biochemistry
<https://facilities.bioc.cam.ac.uk/dna-sequencing-overview/next-generation-sequencing-ngs>

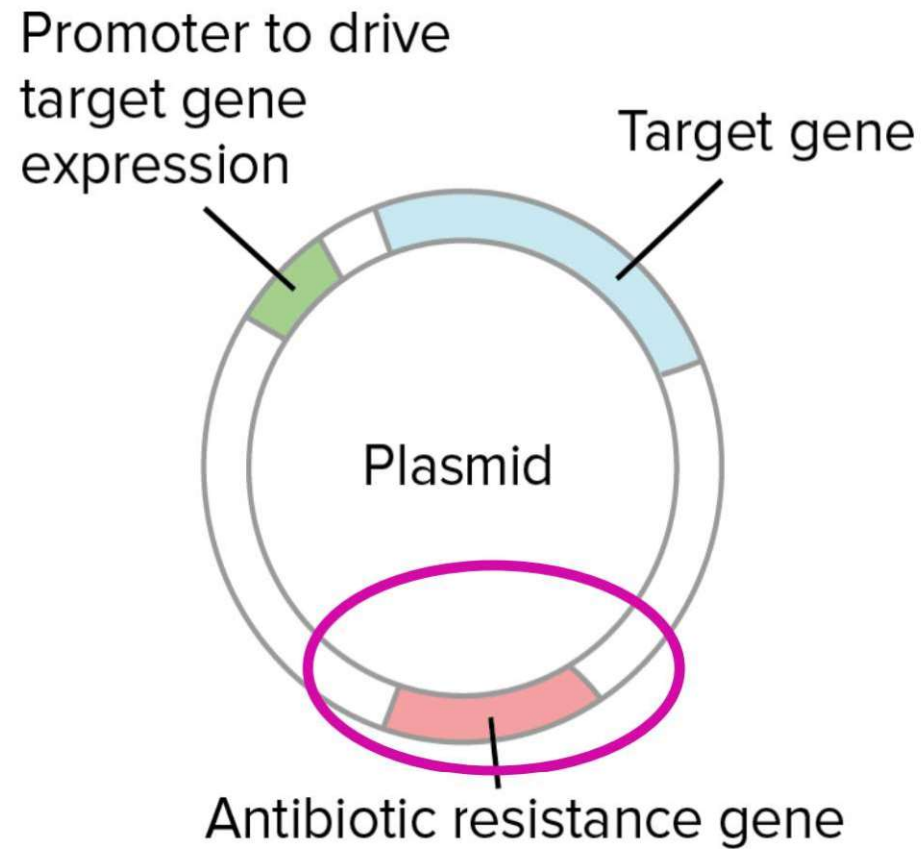
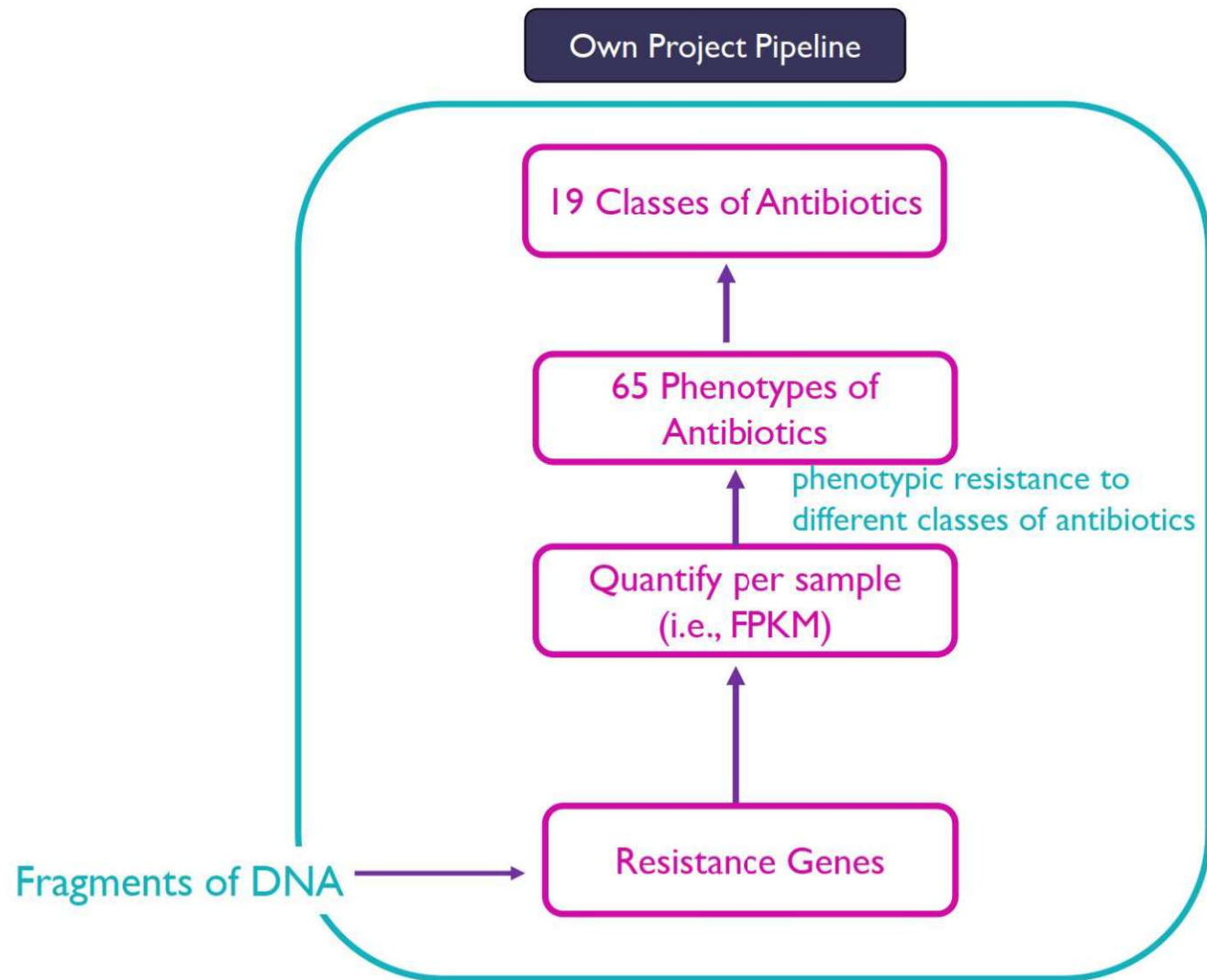


Figure source link: <https://www.khanacademy.org/science/biology/biotech-dna-technology/dna-cloning-tutorial/a/bacterial-transformation-selection>



Whole-genome sequencing output provides a rough estimation of bacteria species abundance

Bacterial species

Own data

Classes of AMR



	A	B	C	D	E	F	G	H	I	J	K
	species_name	genomic_2	genomic_2	genomic_2	genomic_2	genomic_2	genomic_2	genomic_2	genomic_2	genomic_2	genomic_2
2	[Bacillus] caldolyticus	0	0	0	0	0	0	0	0	0	0
3	[Bacillus] clarkii	0	0	0	0	0	0	0	0	0	0
4	[Bacteroides] pectinophilus	3719	419	4260	1882	4858	609	49	613	388	327
5	[Brevibacterium] flavum	0	0	0	0	0	0	0	0	0	0
6	[Clostridium] aerotolerans	0	0	0	0	1	0	0	0	0	0
7	[Clostridium] aminophilum	0	0	0	0	1	1	0	0	1	0
8	[Clostridium] asparagiforme	396	0	260	130	204	101	1	115	53	98
9	[Clostridium] boltea	747	42	702	59	599	160	31	98	67	590
10	[Clostridium] cellulosi	0	0	0	0	0	0	0	0	0	0
11	[Clostridium] citroniae	99	1	110	52	131	6	4	6	3	203
12	[Clostridium] clostridioform	320	52	246	69	323	180	11	194	58	211
13	[Clostridium] cocleatum	0	1	3	0	3	0	1	0	1	0
14	[Clostridium] dakarensis	54	4	4	4	6	2	1	1	4	4
15	[Clostridium] fimetarium	73	1	8	3	5	5	1	4	2	1
16	[Clostridium] glycyrrhizinilyt	851	153	647	275	588	248	57	206	158	103
17	[Clostridium] hiranonis	8	0	3	0	2	0	0	1	1	0
18	[Clostridium] hylemonae	0	0	2	1	0	0	1	7	1	3
19	[Clostridium] innocuum	0	10	8	0	0	34	0	35	17	0
20	[Clostridium] leptum	470	66	308	82	320	547	88	511	144	70
21	[Clostridium] methylpentos	108	4	68	1	94	50	0	1	4	5
22	[Clostridium] paradoxum	11	3	10	3	9	0	0	3	2	7
23	[Clostridium] polysaccharol	0	0	0	0	0	0	0	0	0	0
24	[Clostridium] populeti	1	5	10	2	15	2	0	3	1	3
25	[Clostridium] sporobacter	204	24	121	24	141	132	42	100	64	43
26	[Clostridium] thermocellum	0	0	0	0	0	0	0	0	0	0
27	[Clostridium] thermosaccharo	0	0	0	0	0	0	0	0	0	0
28	[Clostridium] thermosphaera	0	0	0	0	0	0	0	0	0	0
29	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
30	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
31	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
32	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
33	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
34	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
35	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
36	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
37	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
38	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
39	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
40	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
41	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
42	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
43	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
44	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
45	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
46	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
47	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
48	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
49	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
50	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
51	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
52	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
53	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
54	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
55	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
56	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
57	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
58	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
59	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
60	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
61	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
62	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
63	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
64	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
65	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
66	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
67	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
68	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
69	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
70	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
71	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
72	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
73	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
74	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
75	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
76	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
77	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
78	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
79	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
80	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
81	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
82	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
83	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
84	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
85	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
86	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
87	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
88	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
89	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
90	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
91	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
92	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
93	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
94	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
95	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
96	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
97	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
98	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
99	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0
100	[Clostridium] thermosulfidibac	0	0	0	0	0	0	0	0	0	0

Confidential



	A	B	C	D
	country	year	class	fpkm
1	ALB	16	Aminoglycoside	31955.32
2	ALB	17	Aminoglycoside	30062.73
3	ARE	17	Aminoglycoside	67241.45
4	ARE	18	Aminoglycoside	72546.4
5	ARG	17	Aminoglycoside	9997.204
6	ARG	18	Aminoglycoside	17448.68
7	AUS	16	Aminoglycoside	22699.82
8	AUS	17	Aminoglycoside	25087.64
9	AUS	18	Aminoglycoside	18570.99
10	AUT	16	Aminoglycoside	18128.17
11	AUT	17	Aminoglycoside	20671.96
12	AUT	18	Aminoglycoside	20089.6
13	BEL	17	Aminoglycoside	24318.86
14	BEL	18	Aminoglycoside	21786.74
15	BEN	17	Aminoglycoside	76739.81
16	BEN	18	Aminoglycoside	25558.34
17	BFA	17	Aminoglycoside	25680.16
18	BFA	18	Aminoglycoside	25006.62
19	BGD	17	Aminoglycoside	14786.25
20	BGD	18	Aminoglycoside	13299.11
21	BGR	16	Aminoglycoside	32134.56
22	BGR	17	Aminoglycoside	52197.45
23	BGR	18	Aminoglycoside	18528.05



Epidemiological/demographic data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
	new_complete	country	city	latitude	longitude	country_al	Region	income_gr	income	Country_C	SDG_Index	SDG_Index	SDGRegion	Population	Goal1Score	Goal2Score	Goal3Score	Goal4Score	Goal5Score	Goal6Score
1	DTU_2016_1_1	Albania	Tirana	41.32	19.9	ALB	Europe & C	UpperMid	3	ALB	71.02186	highMiddle	71.40243	2877800	91.6375	60.00486	81.09271	95.155	53.57425	75.9076
2	DTU_2016_4_1	Australia	Woden	-35.35	149.09	AUS	East Asia & High	4	AUS	75.57571	high	77.20763	25499881	99.61	58.74013	95.58357	99.4038	81.00025	94.9532	94.9532
3	DTU_2016_9_1	Australia	Melbourne	-37.81	144.96	AUS	East Asia & High	4	AUS	75.57571	high	77.20763	25499881	99.61	58.74013	95.58357	99.4038	81.00025	94.9532	94.9532
4	DTU_2016_14	Austria	Vienna	48.21	16.37	AUT	Europe & C	High	4	AUT	82.08009	high	77.20763	9006400	99.467	73.76713	92.10831	99.20504	83.08375	89.5994
5	DTU_2016_18	Bulgaria	Sofia	42.7	23.32	BGR	Europe & C	UpperMid	3	BGR	73.81489	highMiddle	71.40243	6948445	97.507	66.34725	79.553	67.153	72.472	70.8148
6	DTU_2016_22	Brazil	Belem	-1.45	-48.47	BRA	Latin Amer	UpperMid	3	BRA	71.33911	highMiddle	68.57615	2.13E+08	86.9855	69.89925	77.83657	87.65767	70.29175	86.2456
7	DTU_2016_25	Brazil	Belem	-1.45	-48.47	BRA	Latin Amer	UpperMid	3	BRA	71.33911	highMiddle	68.57615	2.13E+08	86.9855	69.89925	77.83657	87.65767	70.29175	86.2456
8	DTU_2016_28	Botswana	Palapye	-22.54	27.12	BWA	Sub-Sahara	UpperMid	3	BWA	61.92338	lowMiddle	51.92682	2351625	54.784	41.11138	53.73014	89.6733	70.16675	70.5678
9	DTU_2016_32	Canada	Regina	50.45	-104.62	CAN	North Ame	High	4	CAN	79.1554	high	77.20763	37742157	99.5335	67.3625	93.95292	99.58704	82.39025	85.8998
10	DTU_2016_35	Canada	Calgary	51.05	-114.06	CAN	North Ame	High	4	CAN	79.1554	high	77.20763	37742157	99.5335	67.3625	93.95292	99.58704	82.39025	85.8998
11	DTU_2016_40	Canada	Toronto	43.65	-79.39	CAN	North Ame	High	4	CAN	79.1554	high	77.20763	37742157	99.5335	67.3625	93.95292	99.58704	82.39025	85.8998
12	DTU_2016_45	Canada	Ottawa	45.42	-75.69	CAN	North Ame	High	4	CAN	79.1554	high	77.20763	37742157	99.5335	67.3625	93.95292	99.58704	82.39025	85.8998
13	DTU_2016_46	Switzerland	Liebfeld	46.93	7.42	CHE	Europe & C	High	4	CHE	80.0954	high	77.20763	8654618	99.838	70.49525	96.54771	99.61471	88.89701	88.89701
14	DTU_2016_53	China	Guangzhou	23.13	113.26	CHN	East Asia & High	3	CHN	72.06164	highMiddle	65.7492	1.44E+09	98.5655	81.1	81.68677	97.52467	76.20425	68.6428	68.6428
15	DTU_2016_54	Cote d'Ivoire	Abidjan	5.32	-4.02	NA	NA	NA	CIV	57.56248	low	51.92682	26378275	43.7735	61.65763	39.35314	51.82933	39.16025	55.9624	55.9624
16	DTU_2016_58	Czech Rep	Brno	49.19	16.61	CZE	Europe & C	High	4	CZE	81.38791	high	77.20763	7080892	99.795	61.56375	89.7792	98.9533	72.9235	56.5626
17	DTU_2016_61	Germany	Berlin	52.52	13.39	DEU	Europe & C	High	4	DEU	82.4818	high	77.20763	8378394	99.5335	72.44663	93.76762	98.78804	80.70775	84.7016
18	DTU_2016_66	Denmark	Lynetten	55.7	12.62	DNK	Europe & C	High	4	DNK	84.86052	high	77.20763	5792203	99.679	58.85988	94.20269	99.13871	86.78775	90.278
19	DTU_2016_67	Denmark	Avedore	55.63	12.46	DNK	Europe & C	High	4	DNK	84.86052	high	77.20763	5792203	99.679	58.85988	94.20269	99.13871	86.78775	90.278
20	DTU_2016_68	Denmark	Damhusaa	55.64	12.5	DNK	Europe & C	High	4	DNK	84.86052	high	77.20763	5792203	99.679	58.85988	94.20269	99.13871	86.78775	90.278
21	DTU_2016_69	Ecuador	Quito	-0.22	-78.51	ECU	Latin Amer	UpperMid	3	ECU	72.53935	highMiddle	67.164306	85.0345	99.19813	75.56379	97.31	77.29775	75.015	75.015
22	DTU_2016_70	Ecuador	Galapagos	-0.63	-90.36	ECU	Latin Amer	UpperMid	3	ECU	72.53935	highMiddle	68.57615	17643060	85.0345	99.19813	77.56379	97.31	77.29775	75.015
23	DTU_2016_72	Spain	Barcelona	41.38	2.18	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
24	DTU_2016_73	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
25	DTU_2016_74	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
26	DTU_2016_75	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
27	DTU_2016_76	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
28	DTU_2016_77	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
29	DTU_2016_78	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
30	DTU_2016_79	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
31	DTU_2016_80	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
32	DTU_2016_81	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
33	DTU_2016_82	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
34	DTU_2016_83	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
35	DTU_2016_84	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
36	DTU_2016_85	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
37	DTU_2016_86	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
38	DTU_2016_87	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
39	DTU_2016_88	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
40	DTU_2016_89	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
41	DTU_2016_90	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
42	DTU_2016_91	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
43	DTU_2016_92	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
44	DTU_2016_93	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
45	DTU_2016_94	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
46	DTU_2016_95	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
47	DTU_2016_96	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
48	DTU_2016_97	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
49	DTU_2016_98	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
50	DTU_2016_99	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
51	DTU_2016_100	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
52	DTU_2016_101	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
53	DTU_2016_102	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
54	DTU_2016_103	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
55	DTU_2016_104	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
56	DTU_2016_105	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	86.5834
57	DTU_2016_106	Spain	Madrid	40.4168	-3.7038	ESP	Europe & C	High	4	ESP	79.46162	high	77.20763	45647483	98.6075	63.8845	94.12762	96.555	85.81735	

Own data

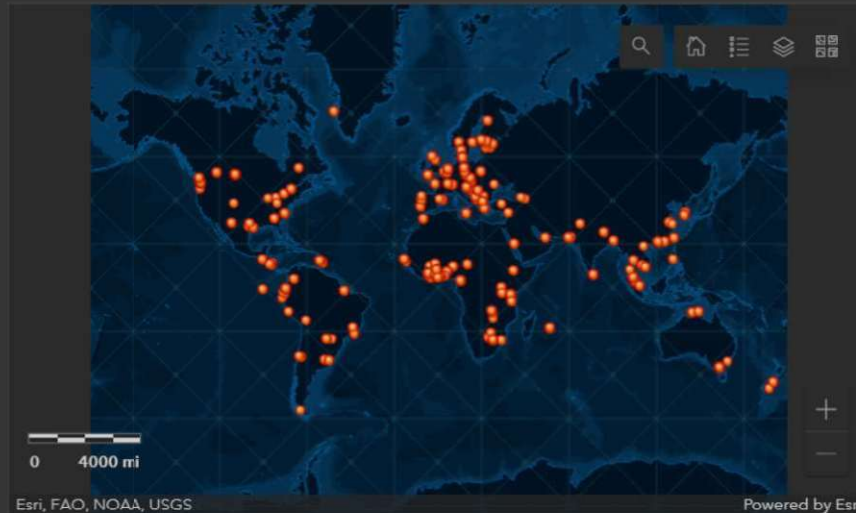
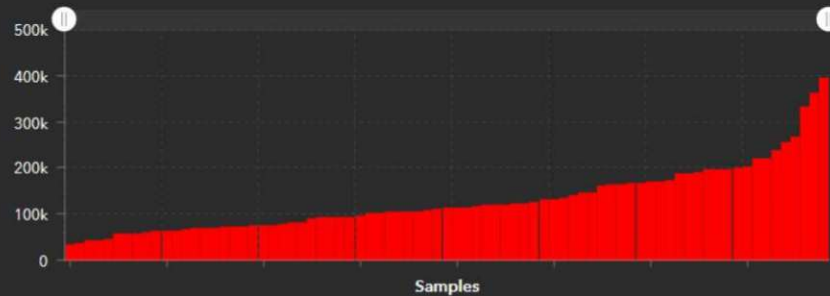
Global Monitoring of Antimicrobial Resistance using Urban Sewage Samples (Phase II): Global Sewage Surveillance Project (...)

Data measures: total counts silva bacterial load, resfinder, and metadata

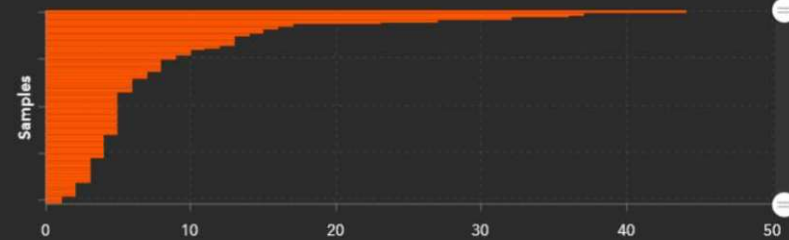
Resfinder Count Per Sample 2016



Bacterial Load Per Sample 2016



Samples per country



About:

This dashboard was developed by Nermin Ghith, Postdoc Researcher at the Research Group for Genomic Epidemiology, Division for Global Surveillance, National Food Institute, Technical University of Denmark (DTU Food).

Go to project StoryMaps:

<https://storymaps.arcgis.com/stories>

Country/City/Municipality/Sample ID

- Albania
Tirana
2016_pil_alb_01
- Albania
Tirana
2017_jun_alb_01
- Albania
Tirana
2017_nov_alb_01
- Argentina
Buenos Aires
2017_nov_arg_01
- Argentina
Buenos Aires
2018_jun_arg_01
- Argentina
Buenos Aires
2018_nov_arg_01

Last update: now

Opportunities and challenges

- Large collection of samples (and countries)
- Public interest and international network of collaborators
- WGS technology but metagenomics data (bacterial taxa)
- Too many zeroes (up to 73-80% at the species and gene levels)
- Aggregate resistance genes into phenotypes (antibiotics) and phenotypes into classes of AMR (zeroes are around 6%)
- Selection and interpretation of data analyses



Methods

- The Global Sewage Surveillance project consortium collects untreated sewage samples to assess the global burden of AMR.
- Site-specific samples (560 and 774) were collected between 2016 and 2018 and came from 233 cities (102 countries).
- Samples were sequenced to obtain information on 19 classes of antimicrobial resistance.
- The read counts per class of AMR were transformed using the centered log-ratio (clr) to obtain the outcome measures.



Methods

- Applying **random effect models**, the contextual effect of the city was quantified using the variance partition confident (VPC) by adding the random effects of the city in the analyses.
- The model **variance** was estimated as the median of the posterior distribution obtained by the Markov Chain Monte Carlo (MCMC) method.
- **SDG indices** (determinants) were added into the models to evaluate the study aim.
- Data were extracted from the World Bank and the Global Burden of Disease data portals.
- Commercial data bought from GIDEON and IQVIA, among other resources.
- Use proportional change in variance (**PCV**) to assess the effect of the determinants (each in a separate model)



Methods

Determinants in the models

- SDG global index: covers all the 17 goals
- Universal Health Coverage (UHC)
- Vaccine Cover
- WaSH Mortality: * Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (exposure to unsafe Water, Sanitation and Hygiene for All (WASH))
- Access to clean water
- GBD-SDG health: index composed of 41 health indicators



Findings

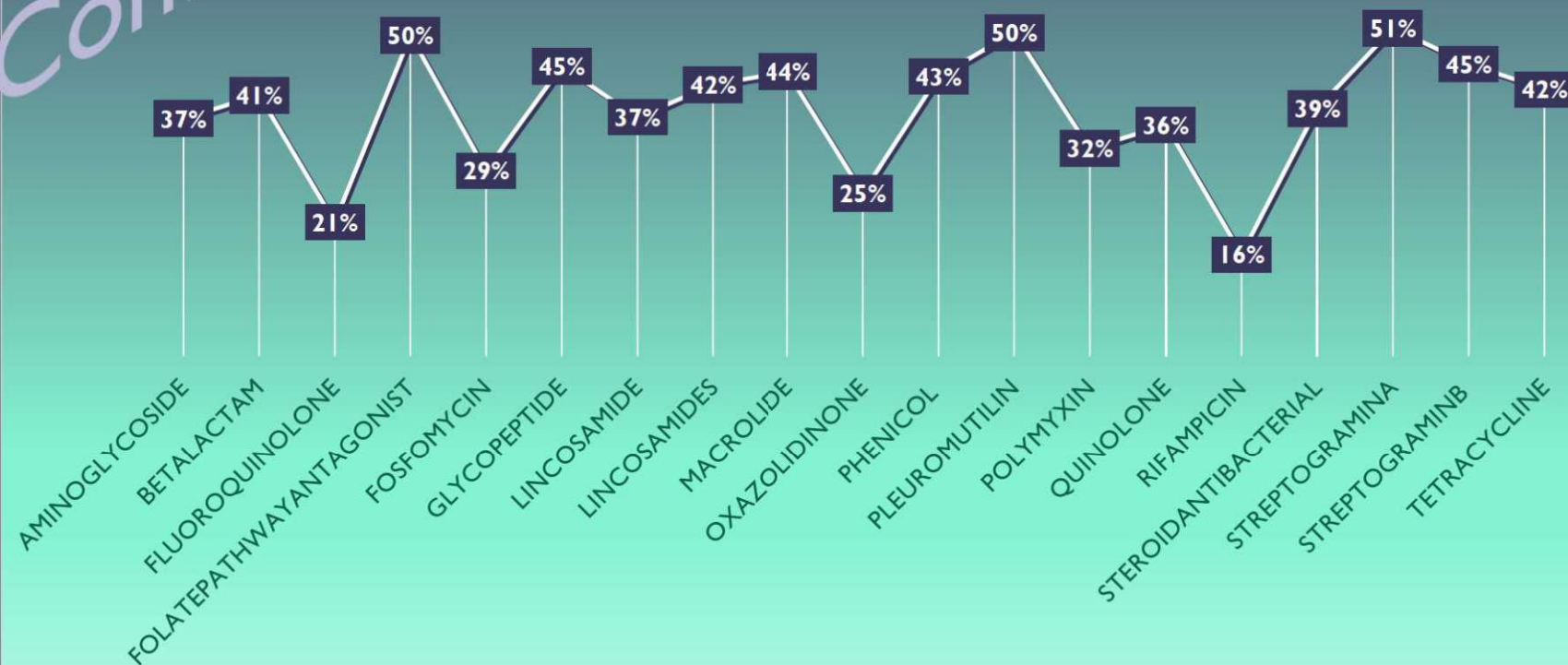
- Adding the **SDG** global index into the random intercept models explained around **6%-55%** of the between city variation in AMR .
- Similar findings obtained by adding the indices on Universal Health Coverage (**UHC**), Vaccine Cover, **WaSH Mortality** and access to clean **water** indicators into the models.
- In all the models, the largest reduction in between **city variation in the burden** of AMR occurred for **folate pathway antagonist**, **Phenicol**, and **Tetracycline classes**.
- The least reduction was for **polymyxin** and **quinolone** classes.



Own data

Confidential

FIG. 1. EXPLAINED VARIANCE:
RANDOM EFFECT MODEL WITH
SDG GLOBAL INDEX SCORE*



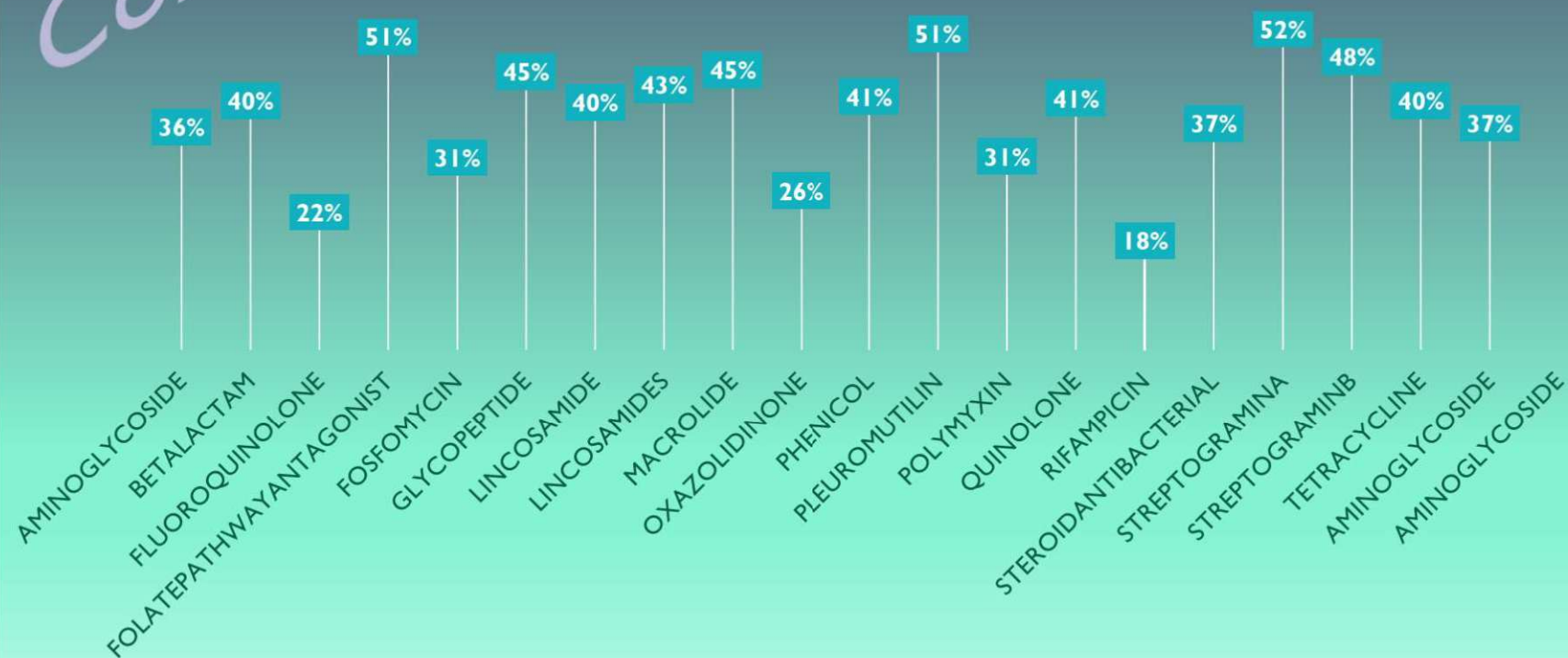
*The index covers all the SDGs 17 goals



Own data

Confidential

FIG. 2. EXPLAINED VARIANCE:
RANDOM EFFECT MODEL WITH
SDG **HEALTH INDEX***



*The index covers 41 health indicators (health index)



Own data

FIG. 3. EXPLAINED VARIANCE:
RANDOM EFFECT MODELS WITH INDICES ON SANITATION, UNIVERSAL HEALTH
COVERAGE, VACCINE COVER, WASH* MORTALITY, AND ACCESS TO CLEAN WATER



* Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (exposure to unsafe Water, Sanitation and Hygiene for All (WASH))





Sustainability Development Goals and biomarkers of Antimicrobial Resistance: differential abundance analyses

Nermin Ghith, Hannah-Marie Martiny, Patrick Munk, Amalia Bogri, and Frank Møller Aarestrup
Research Group of Genomic Epidemiology, Technical University of Denmark (DTU)



Research group of Genomic Epidemiology
Technical University of Denmark
Mobile: +4523190249
nergh@dtu.dk, hmar344@dtu.dk
<https://www.linkedin.com/company/nermin-ghith-50455316/>

Revised abstract

Background:

Inequality in global health outcomes such as antimicrobial resistance (AMR) is shaped by macro determinants of health within and between countries.

Therefore, this study aims to assess the potential relationship between the country's development level - progress towards achieving the sustainability development goals (SDG) and the burden of AMR.

The study identifies resistance genes, phenotypes (antibiotics), classes of antimicrobial resistance and bacterial taxa that are differentially abundant between the sampling sites within countries, and compliance with SDGs goals that might explain variation in the global burden of AMR between countries.



Methods:

The Global Sewage Surveillance project consortium collects untreated sewage samples to assess the global burden of AMR. Samples were sequenced to obtain data on the microbial compositions within site-country covering information on resistance genes and bacterial taxa. Samples were collected between 2016 and 2018 and came from 233 cities (102 countries).

A combination of models is developed to describe and characterize the data on AMR in association with SDG indices and goals. A number of SDG indices were added to the analyses to evaluate the study aim. Data on SDGs indices were extracted from the World Bank and the Global Burden of Disease data portals.

Results:

The microbial communities across sites differ from one another, which are attributed to site-country development levels (e.g., measured with SDG index and goals). The study identified a number of differentially abundant taxa at different levels of the bacterial taxonomy as well as differentially abundant genes, phenotypes and classes of AMR.

Conclusions:

Further interventions focusing on improving adherence to SDG goals have the potential for elevating inequality in the global burden of AMR. Yet, a complementary assessment for additional sampling locations is required to get data that are more granular on the global and local burden of AMR and to determine the most appropriate course of action.

Methods

Metagenomics samples were sequenced to obtain information on 19 classes of antimicrobial resistance, 65 phenotypes, 562 resistance genes and bacterial taxa.

To assess the study aim, the findings were triangulated using the wilcoxon-mann-whitney test (WMW), negative binomial models (NB), random effect models, compositional data analysis, and analysis of compositions of microbiomes with bias correction (ANCOM-BC or ABC).

Applying random effect models, the contextual effect of the city was quantified using the variance partition coefficient (VPC) by adding the random effects of the city in the analyses.

A number of SDG indices were added into the models to evaluate the study aim. Data on SDGs indices were extracted from the World Bank and the Global Burden of Disease data portals. The model variance was estimated as the median of the posterior distribution obtained by the Markov Chain Monte Carlo (MCMC) method.

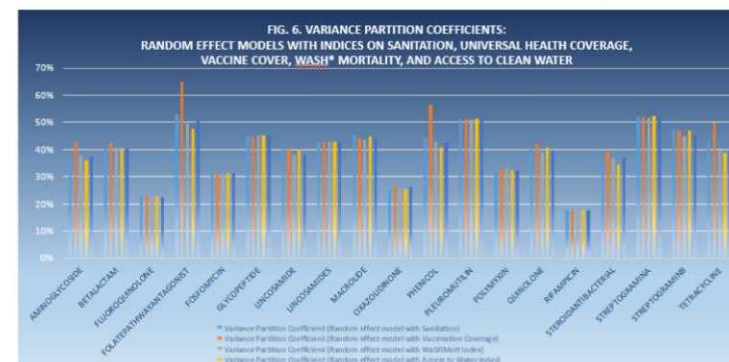
The work is supported by The Novo Nordisk Foundation (NNF16OC0021856: Global Surveillance of Antimicrobial Resistance). Photos credit: unsplash website



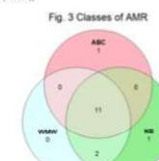
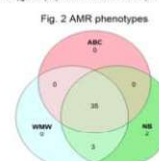
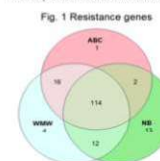
*The index covers all the 17 goals



*The index covers 41 health indicators



*Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (exposure to unsafe Water, Sanitation and hygiene for All (WASH))



Results

The microbial communities across sites differ from one another, which is attributed to site-country development levels (e.g., measured with SDG index and goals). The study identified 114 resistance genes, 35 phenotypes and 11 classes (Figs. 1-3) that are differentially abundant between countries.

Classes of AMR that have the smallest p-values across the random effect model analyses are: Aminoglycoside, Folate pathway antagonist, Lincosamide, Macrolide, Oxazolidinone, Phenicol, Pleuromutilin, Polymyxin, Streptogramin-A, Streptogramin-B, and Tetracycline.

Adding the SDG global index and SDG health scores into the random intercept models explained around 18%-52% of the between city variation in total AMR (Figs 4 and 5). Similar findings obtained by adding the indices on Sanitation, Universal Health Coverage (UHC), Vaccine Cover, WASH Mortality, and Access to water individually into the models. The variance partition coefficients (VPC) values are between 18%-65% (Fig 6).

In random effect models (Figs 4-6), considering the SDG indices, the largest reduction in between city variation in the burden of AMR occurred for folate pathway antagonist, pleuromutilin and streptogramin-A classes, and the least reduction (still quantitatively large) was for fluoroquinolone, oxazolidinone and rifampicin classes.

Conclusions

Actions focusing on sustained compliance with the SDG goals have the potential for elevating inequality in the global burden of AMR. There is a need for additional data that cover more sampling locations to develop more comprehensive global monitoring of AMR, and to determine the most appropriate course of action.

Conclusion

- Further **interventions** focusing on improving adherence to SDG goals have the potential for elevating the global burden of AMR.
- Yet, a complementary assessment for **additional locations** is required to get more granular data on the global burden of AMR and to determine the most appropriate course of action.
- Consider other **analytical** frameworks: e.g., log linear transformation of FPKM and offset for the library size



Teaching: New trends in surveillance

Surveillance and Epidemiology of infectious Diseases

MSc in Bioinformatics – master track in Infectious Disease Informatics



```
Sewage II 2024 - RStudio
Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Data processing bacteria taxa.Rmd SDGsVIZ.qmd SDGs2024_Q2.2_AllData_After_20240729 SDG.Rmd
Knit on Save Knit Run
Source Visual Outline
1 ----
2 title: "Data processing class Pheno u90 total"
3 author: "Nermin Ghith"
4 date: "2024-04-17"
5 output: html_document
6 ----
7
8 # view the data
9
10 {r setup, include=FALSE}
11 knitr::opts_chunk$set(echo = TRUE)
12
13 # List of file names
14 # file_names <- c("bacteriaLoadPerSample", "resfinder.counts.phenotype", "resfinder.counts.uc90",
15 # "resfinder.fpk.class", "resfinder.fpk.phenotype",
16 # "resfinder.fpk.uc90", "total.counts.silva+resfinder")
17 #
18 # Loop through each file name
19 # for (file in file_names) {
20 #   # Construct the file path
21 #   file_path <- paste0("c:/Users/nermi/Dropbox/DTU/Global determinants of AMR/Sewage II
22 # 2024/Data/kma_quantification/products/", file, ".txt")
23 #
24 #   # Read the file and assign it to a unique variable in the global environment
25 #   assign(file, read.delim(file_path))
26 # }
27 # # manually view each data frame in RStudio with view(resfinder.counts.phenotype), etc.
28
29
30
31
32 # Set the path to the data directory
33 directory_path <- "c:/Users/nermi/Dropbox/DTU/Global determinants of AMR/Sewage II
```

